Awesome AI4Bio (Artificial Intelligence for Biology)

一个精选的 **AI for Biology** 领域文献清单，持续更新中。

Foundation Models

基于大规模生物组学数据预训练的基础模型。

Year / Venue	Paper	Tags
2025	**scGPT: End-to-End Protocol for Fine-Tuned Retinal Cell Type Annotation** [paper] [GitHub]	`Single-cell omics`
2025	**CELLama: Foundation Model for Single Cell and Spatial Transcriptomics by Cell Embedding Leveraging Language Model Abilities** [paper]	`Single-cell omics`, `Spatial omics`
2025	**scMOBA: A Conversational Single-Cell Multi-Omics Brain Agent Across Species** [paper]	`Single-cell omics`, `Multi-omics`
2024 *Nature Methods*	**scGPT: Toward Building a Foundation Model for Single-Cell Multi-Omics Using Generative AI** [paper] [GitHub]	`Single-cell omics`, `Multi-omics`
2024 *Genome Biology*	**scmFormer: Integrates Large-Scale Single-Cell Proteomics and Transcriptomics Data by Multi-Task Transformer** [paper]	`Single-cell omics`, `Multi-omics`

AI Agents

具备自主规划、工具调用与反思迭代能力的 LLM 智能体，完成生物数据分析任务。

Year / Venue	Paper	Tags
2026 *Nature*	**Accelerating Scientific Discovery with Co-Scientist** [paper]	`Cross-Modal` , `Scientific Discovery`
2026 *Nature*	**An AI System to Help Scientists Write Expert-Level Empirical Software (ERA)** [paper]	`Tools & Infrastructure` , `Pipeline Optimization`
2026 *Nature*	**A Multi-Agent System for Automating Scientific Discovery (Robin)** [paper]	`Cross-Modal` , `Scientific Discovery`
2026 *Nature Methods*	**CellVoyager: AI CompBio Agent Generates New Insights by Autonomously Analyzing Biological Data** [paper]	`Single-cell omics` , `Scientific Discovery`
2026 *Nature Biomedical Engineering*	**BioMedAgent: Empowering AI Data Scientists Using a Multi-Agent LLM Framework with Self-Evolving Capabilities for Autonomous Tool-Aware Biomedical Data Analyses** [paper] [GitHub]	`Cross-Modal` , `Pipeline Optimization`
2026 *Genome Biology*	**Benchmarking LLM-Based Agents for Single-Cell Omics Analysis** [paper]	`Single-cell omics`
2026 *arxiv*	**CELLAGENT: LLM-Driven Multi-Agent Framework for Natural Language-Based Single-Cell Analysis** [paper] [GitHub]	`Single-cell omics` , `Pipeline Optimization`
2026 *biorxiv*	**CellAtria: An Agentic AI Framework for Ingestion and Standardization of Single-Cell RNA-seq Data Analysis**	`Single-cell omics` , `Pipeline Optimization`

Year / Venue	Paper	Tags
	[paper] [GitHub]	
2026 *biorxiv*	**PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery** [paper] [Website]	`Cross-Modal` , `Scientific Discovery`
2025 *Nature Communications*	**CASSIA: A Multi-Agent Large Language Model for Automated and Interpretable Cell Annotation** [paper]	`Single-cell omics` , `Pipeline Optimization`
2025 *Briefings in Bioinformatics*	**ANNOAGENT: A Language Agent for Single-Cell Automatic Annotation** [paper]	`Single-cell omics`
2025 *biorxiv*	**scAgent: Universal Single-Cell Annotation via a LLM Agent** [paper]	`Single-cell omics` , `Pipeline Optimization`
2025 *biorxiv*	**Multi-Agent AI Enables Evidence-Based Cell Annotation in Single-Cell Transcriptomics** [paper] [GitHub]	`Single-cell omics` , `Pipeline Optimization`
2025 *biorxiv*	**InstructCell: A Multi-Modal AI Copilot for Single-Cell Analysis with Instruction Following** [paper] [GitHub]	`Single-cell omics` , `Pipeline Optimization`
2025 *biorxiv*	**CELLFORGE: Agentic Design of Virtual Cell Models** [paper]	`Single-cell omics` , `Scientific Discovery`
2025 *biorxiv*	**BioDiscoveryAgent: An AI Agent for Designing Genetic Perturbation Experiments** [paper]	`Single-cell omics` , `Scientific Discovery`
2024 *Nature Methods*	**Omega: Harnessing the Power of Large Language Models for Bioimage Analysis** [paper]	`Bioimage` , `Single-cell omics`
2024 *Nature Methods*	**BioImage.IO Chatbot: A Community-Driven AI Assistant for Integrative**	`Bioimage`

Year / Venue	Paper	Tags
	Computational Bioimaging [paper]	
2024 *Genome Biology*	**Assessing GPT-4 for Cell Type Annotation in Single-Cell RNA-seq Analysis** [paper]	`Single-cell omics`
2024 *Small Science*	**AutoBA: An AI Agent for Fully Automated Multi-Omic Analyses** [paper] [GitHub]	`Single-cell omics`, `Pipeline Optimization`

Benchmarks

该领域的系统性综述与基准评测研究。

Year / Venue	Paper	Tags
2025 *Advanced Science*	**Artificial Intelligence Revolution in Transcriptomics: From Single Cells to Spatial Atlases** [paper]	`Single-cell omics`, `Spatial omics`
2022 *Nature Methods*	**Benchmarking Atlas-Level Data Integration in Single-Cell Genomics** [paper]	`Single-cell omics`

Tools & Infrastructure

广泛使用的生物信息学工具与计算基础设施。

Year / Venue	Paper	Tags
2018 *Genome Biology*	**SCANPY: Large-Scale Single-Cell Gene Expression Data Analysis** [paper] [GitHub]	`Single-cell omics`, `Tools & Infrastructure`

Tags 说明

Tag	Description
`Single-cell omics`	应用于单细胞组学数据分析，包括 scRNA-seq、scATAC-seq、scMulti-omics 等
`Spatial omics`	应用于空间组学数据分析，包括空间转录组、空间蛋白质组等
`Multi-omics`	模型设计整合了多种组学层次（如转录组 + 蛋白组 + 表观组）
`Bioimage`	应用于生物医学图像分析，包括细胞图像、显微图像等
`Cross-Modal`	Agent 能力跨越多个模态或领域，非单一组学任务
`Scientific Discovery`	核心能力为自主提出科学假设、设计实验方案、推动生物学新发现
`Pipeline Optimization`	核心能力为自主构建、优化、执行端到端生物信息学分析流程
`Tools & Infrastructure`	经典生物信息学工具或计算基础设施，具有广泛的应用价值